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Grade 11 Geography

3.Inclined Rock Strata

Learning Guide

🟧 1. Why Do Rock Strata Tilt?

- Rock layers are **originally deposited horizontally**.
- Over time, they become **inclined (tilted)** due to internal Earth forces.

Main Causes:

- **Weight of sediments:**
  - Heavy layers build up over time → pressure causes tilting.
- **Volcanic intrusions:**
  - Magma pushes from below → forces rock layers upward.
- **Tectonic movement:**
  - Movement of plates → **folding (bending)** or **faulting (breaking)** of rock layers.

📌 *Key Idea:* Inclined strata = **evidence of internal Earth forces at work**

🟫 2. Topography of Inclined Strata

- Rock layers consist of **alternating hard and soft rocks**.
- When exposed to **weathering and erosion**, they form **distinct landforms**.

What forms?

- **Asymmetrical ridges** (uneven slopes on each side)

Two Main Slopes:

- **Dip slope:**
  - Forms on **hard, resistant rock**
  - Erodes slowly → becomes **gentle slope**
- **Scarp slope:**
  - Forms on **soft, less resistant rock**
  - Erodes quickly → becomes **steep slope**

📌 *Important:* Difference in resistance = different slope shapes

🟦 3. Understanding Dip vs Scarp Slope (CORE CONCEPT)

- **Dip slope:**
  - Follows the **angle of rock layers**
  - Smooth and gradual
  - Easier for farming and settlement
- **Scarp slope:**
  - Cuts **across rock layers**
  - Very steep and rugged
  - More prone to erosion and rockfalls

📌 *Exam Tip:*

Dip = gentle (hard rock)

Scarp = steep (soft rock)

⛰️ 4. Types of Inclined Landforms (By Dip Angle)

🟧 Cuesta (10°–25°)

- Very **gentle dip slope**
- Wide and stretched landform
- Most common and easiest to recognise

🟫 Homoclinal Ridge (25°–45°)

- **Moderate slope angle**
- More balanced between steep and gentle sides
- Often used as a **general term** for these ridges

🟥 Hogsback (>45°)

- Very **steep dip slope**
- Narrow, sharp ridge
- Harder to erode → more dramatic landscape

📌 *Key Idea:*

As dip angle increases → slope becomes steeper and sharper

🔄 5. Homoclinal Shifting (Scarp Retreat in Inclined Strata)

- The **scarp slope is gradually eroded backward** over time.

What happens step-by-step:

1. Soft rock at the **base of the scarp** is eroded
2. Upper rock loses support → collapses
3. Entire slope shifts **backward**

Results:

- Ridge moves in direction of **dip slope**
- Ridge becomes:
  - **Narrower**

    - **Lower**
- Happens in **all inclined landforms**

📌 *Important:* Shape stays similar, but position changes

🟩 6. Cuesta Basins (Sagging Landforms)

- Formed when **magma sinks downward (lopolith intrusion)**

Process:

- Rock layers **sink inward (sagging)**
- Creates a **bowl-shaped depression**
- Erosion forms **circular ridges**

Slope Direction:

- **Dip slope → faces inward**
- **Scarp slope → faces outward**

📍 Example: **Bushveld Igneous Complex**

📌 *Memory Tip:*

Basin = bowl = slopes face inside

🟪 7. Cuesta Domes (Uplifted Landforms)

- Formed when **magma pushes upward (laccolith/batholith)**

Process:

- Rock layers are **uplifted and pushed outward**
- Erosion creates a **circular dome shape**

Slope Direction:

- **Dip slope → faces outward**
- **Scarp slope → faces inward**

📌 *Memory Tip:*

Dome = pushed up = slopes face outside

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